

Wiring Guide

Document Type: Circuit Wiring Reference

Version: 1.0

Prepared For: Assembly & Testing

Overview

This document explains the wiring layout and connections between all major components: **Battery, Buck Converter, Dual BTS7960 Motor Drivers, and ESCs.** It's designed to simplify setup, ensure consistent grounding, and avoid polarity mistakes.

Circuit Connection Table

Component	Pin / Wire	Connected To	Purpose / Description
 Battery	Positive (+)	Buck Converter Input (+)	Powers the 5V buck converter
 Battery	Positive (+)	Left & Right BTS7960 B+	Provides main motor power input
 Battery	Negative (-)	Common Ground Bus	Creates unified ground reference
 Buck Converter	Output (+5V)	Both BTS7960 5V pins (grouped of 3)	Supplies logic power to both drivers
 Buck Converter	Output GND	Common Ground Bus	Logic ground connection
 Left BTS7960	GND	Common Ground Bus	Ground for motor driver
 Left BTS7960	B+	Battery Positive (+)	Motor power input
 Left BTS7960	B-	Common Ground Bus	Power ground
 Right BTS7960	GND	Common Ground Bus	Ground for motor driver

 Right BTS7960	B+	Battery Positive (+)	Motor power input
 Right BTS7960	B-	Common Ground Bus	Power ground
 Both BTS7960	5V Pins	Buck Converter +5V Output	Shared logic power supply
 Left ESC	Battery Red	BTS7960 5V Line (common +5V)	ESC logic power input
 Left ESC	Battery Black	Common Ground Bus	ESC ground input
 Right ESC	Battery Red	BTS7960 5V Line (common +5V)	ESC logic power input
 Right ESC	Battery Black	Common Ground Bus	ESC ground input
 Left ESC	Motor Red	Left BTS7960 LPWM	Motor signal (left wheel)
 Left ESC	Motor Black	Left BTS7960 RPWM	Motor signal (left wheel)
 Right ESC	Motor Red	Right BTS7960 LPWM	Motor signal (right wheel)
 Right ESC	Motor Black	Right BTS7960 RPWM	Motor signal (right wheel)
 Each ESC	Switch Wires	Shorted Together	Always ON (bypassed switch)

Recommended Wiring Order

1. **Step 1:** Connect all **grounds (-)** to the common ground bus.
2. **Step 2:** Connect **battery +** to both BTS7960 B+ and buck converter input.
3. **Step 3:** Wire the **buck converter output** to both BTS7960 5V pins.
4. **Step 4:** Connect **ESC logic power (red/black)** to the same 5V and ground lines.
5. **Step 5:** Attach **LPWM/RPWM** lines between ESCs and respective BTS7960.
6. **Step 6:** Short the ESC switch wires to **keep power always ON**.
7. **Step 7:** Double-check all polarities before connecting the main battery.

Safety & Testing Notes

- Always verify voltage from the **buck converter** before connecting logic lines (should be exactly **5V**).
- Use thick enough wires for **motor power (16–18 AWG)** to prevent heating.
- Do a **continuity check** for all grounds before first power-up.
- Start with **low PWM speed** during testing.
- Avoid connecting or disconnecting components while the battery is live.



Component Specifications

Below are the recommended specifications and details for each part used in this setup. These ensure proper power handling, compatibility, and safety during operation.

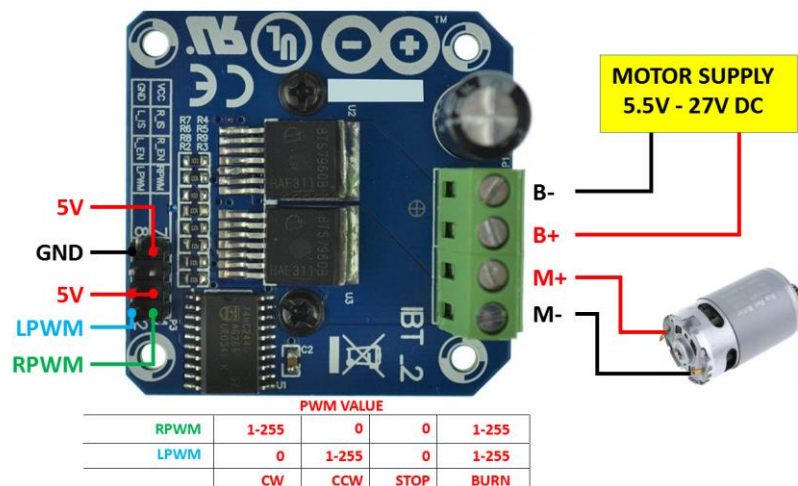
Battery

Specification	Details
Type	Li-ion / LiPo (Rechargeable)
Voltage	11.1 V (3S) or 12 V DC
Capacity	2200 mAh – 5000 mAh (depending on runtime needs)
Discharge Rating (C-rate)	≥ 25C (for stable motor current)
Connector Type	XT60 or DC barrel plug
Purpose	Main power source for motors and logic (through buck converter)

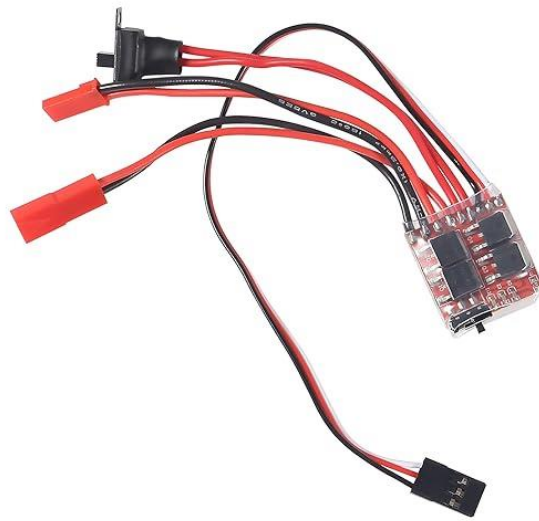
Buck Converter (Step-Down Module)



BTS7960 Motor Driver (x2)



⚡ ESC (Electronic Speed Controller, x2)



💡 Wiring & Connectors

Specification	Details
Power Wires	16–18 AWG (silicone-insulated, flexible)
Signal Wires	22–24 AWG
Connectors	Dupont / JST / XT60 as needed
Protection	Heat-shrink tubing, cable sleeves
Purpose	Safe and organized electrical connections